

TempoRAG-KG: Temporal Knowledge Graph-Augmented Retrieval for Multi-Hop Question Answering

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Abstract

Knowledge Graph-augmented Retrieval (KG-RAG) systems have demonstrated strong performance on multi-hop question answering by structuring retrieved facts as interconnected triples. However, existing systems—including the recent state-of-the-art KG²RAG—assume a static snapshot of knowledge and fail to account for *temporal validity*: the reality that facts are only true within specific time windows. In this paper, we present **TempoRAG-KG**, a temporally-aware KG-RAG framework that extends KG²RAG with two contributions: (1) explicit `[valid_from, valid_to]` validity intervals attached to every KG triple, enabling deterministic temporal filtering during retrieval, and (2) a Graph-of-Graph (GoG)-based fill-in mechanism for handling incomplete KG facts. We first conduct a systematic analysis of temporal failure modes in KG-RAG, categorizing failures into four types and measuring their distribution across HotpotQA and MuSiQue. Our exploratory data analysis reveals that 35.0% of HotpotQA and 38.3% of MuSiQue development questions contain temporal expressions, with prevalence increasing from 30.2% in 2-hop to 46.7% in 4-hop questions—suggesting temporal validity becomes increasingly critical as reasoning complexity grows. We hypothesize that TempoRAG-KG will improve F1 over the KG²RAG baseline, with gains concentrated in temporally-sensitive question subsets.

1 Introduction

Retrieval-Augmented Generation (RAG) has emerged as the dominant paradigm for grounding large language model (LLM) responses in external factual knowledge (Lewis et al., 2020). Recent advances have moved beyond flat vector retrieval toward structured graph-based approaches, where Knowledge Graphs (KGs) provide fact-level relationships between retrieved chunks. The state-of-the-art KG²RAG (Zhu et al., 2025) demonstrates

that organizing retrieved evidence through KG-guided expansion and Maximum Spanning Tree context organization yields substantial improvements over prior RAG baselines on multi-hop question answering benchmarks such as HotpotQA (Yang et al., 2018).

However, a fundamental limitation remains unaddressed: **all existing KG-RAG systems, including KG²RAG, assume a static knowledge graph in which every triple is treated as universally and permanently valid.** In reality, facts are only true within specific time periods—Tim Cook *became* CEO of Apple in 2011, implying Steve Jobs held that role before him. When a user asks “*Who was the CEO of Apple in 2005?*”, a system without temporal awareness retrieves both facts simultaneously and either hallucinates or selects arbitrarily. We term this class of errors *temporal failure modes*. Despite the prevalence of such queries—our exploratory data analysis (EDA) reveals that 35.0% of HotpotQA and 38.3% of MuSiQue development questions contain temporal expressions—no existing KG-RAG framework systematically addresses temporal validity at the triple level on general-domain unstructured text.

To address this gap, we propose **TempoRAG-KG**, a framework that extends KG²RAG with two targeted components. First, we attach explicit `[valid_from, valid_to]` validity intervals to every extracted triple during offline KG construction, and apply deterministic temporal filtering prior to graph expansion. Second, we integrate a GoG-based fill-in mechanism (He et al., 2024) to recover answers when the constructed KG is incomplete. Crucially, we precede system development with a *systematic diagnostic study* that categorizes temporal failure modes into four types and quantifies their distribution across two established multi-hop QA benchmarks.

We anticipate several key findings from our experiments. First, stale-fact retrieval will constitute

the majority of temporal errors in KG²RAG, as explicit year mention (17.8% of HotpotQA) frequently surfaces conflicting facts from different eras. Second, temporal prevalence will escalate with reasoning complexity—from 30.2% in 2-hop MuSiQue questions to 46.7% in 4-hop—suggesting that temporal awareness is disproportionately important for complex multi-hop chains. Third, LLM-based temporal extraction will achieve high precision on explicit date expressions but degrade on implicit and conflicting temporal references.

The contributions of this work are:

1. **Temporal Failure Taxonomy:** A systematic categorization of four temporal failure modes in KG-RAG systems, with empirical distribution analysis across HotpotQA and MuSiQue.
2. **TempoRAG-KG Framework:** A temporally-aware KG-RAG pipeline that attaches explicit validity intervals per triple and applies temporal filtering prior to graph expansion, extending KG²RAG (Zhu et al., 2025).
3. **Temporal Extraction Study:** An empirical evaluation of LLM-based temporal validity extraction from unstructured text, characterizing accuracy across three linguistic pattern types.

2 Related Work

2.1 KG-Augmented Retrieval-Augmented Generation

Early RAG systems (Lewis et al., 2020) rely on dense passage retrieval to retrieve isolated text chunks, ignoring structural relationships between facts. Graph-based RAG approaches address this by incorporating KG structure into retrieval. KG²RAG (Zhu et al., 2025) proposes KG-guided chunk expansion using seed retrieval followed by a KG-based chunk organization process via Maximum Spanning Tree, achieving strong performance on HotpotQA. GEAR (Ma et al., 2025) introduces efficient beam-search graph traversal that replaces expensive LLM calls at each hop with cosine similarity scoring, substantially reducing inference cost. K-RagRec (Wang et al., 2025) applies KG-guided RAG to recommendation, achieving up to 93% hallucination reduction. Despite these advances, **none of these systems attach temporal validity metadata to KG triples**, treating all retrieved facts as equally and permanently valid.

2.2 Temporal Knowledge Graph Reasoning

Temporal KG reasoning has been studied extensively over pre-built structured KGs such as Wikidata and ICEWS. TempAgent (Xiong et al., 2025) introduces temporal filtering tools for an LLM agent, but requires a pre-existing structured temporal KG and cannot operate over general-domain unstructured text. EvoReasoner (Lin et al., 2025) proposes a unified framework combining multi-hop temporal reasoning with incremental KG evolution from documents via confidence-based contradiction resolution and temporal trend tracking—the closest existing work to TempoRAG-KG. However, EvoReasoner tracks temporal evolution *implicitly* through confidence scores rather than explicit `[valid_from, valid_to]` intervals, which prevents deterministic point-in-time querying. Furthermore, EvoReasoner is evaluated on structured TKGQA benchmarks (MultiTQ, CronQuestions) rather than general-domain multi-hop QA datasets. **Our work is the first to combine explicit triple-level temporal validity tagging with general-domain KG-RAG and evaluate on HotpotQA and MuSiQue.**

2.3 KG Construction from Unstructured Text

Automatic KG construction enables RAG systems to operate without pre-built KGs. HRGraph (Mahbubul Alam et al., 2024) and STINMatch (Jin et al., 2023) use LLM prompting to extract subject-relation-object triples from raw text, while TOBUGraph (Acharya et al., 2024) demonstrates production-scale construction. GoG (He et al., 2024) addresses KG incompleteness by allowing LLMs to fill in missing facts from parametric knowledge when graph traversal reaches dead ends. However, **none of these approaches extract or store temporal validity intervals alongside triples**, losing the temporal context necessary for time-sensitive queries. TempoRAG-KG extends LLM-based extraction to additionally infer `valid_from` and `valid_to` for each triple.

Gap Summary. No existing system simultaneously (1) automatically constructs a KG from general-domain unstructured text, (2) attaches explicit per-triple temporal validity intervals, and (3) evaluates on multi-hop QA benchmarks. TempoRAG-KG fills this gap.

3 Research Questions

This work is organized around three research questions:

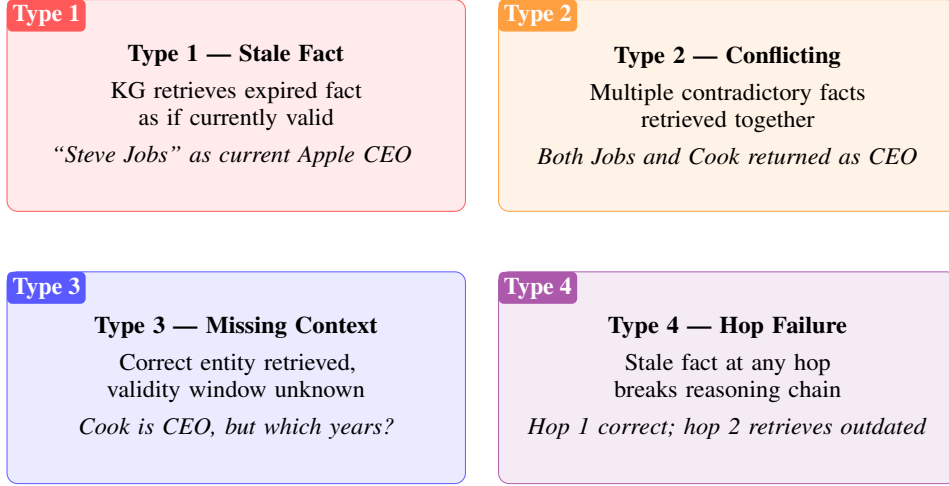


Figure 1: Four types of temporal failure modes in KG-RAG systems, identified through qualitative analysis of KG²RAG errors on the HotpotQA temporal subset.

RQ1 *What types of temporal failures occur in KG-RAG systems when answering time-sensitive multi-hop questions, and how are they distributed across HotpotQA and MuSiQue?*

IV: Question type (temporal vs. non-temporal), hop count. **DV:** Failure type distribution, failure rate per category.

H1: Stale-fact retrieval (Type 1) will be the most frequent failure mode, and failure rate will increase with hop count in MuSiQue.

RQ2 *To what extent does explicit triple-level temporal validity tagging improve retrieval precision and answer F1 on temporally-sensitive questions compared to the KG²RAG baseline?*

IV: Presence of temporal tagging and filtering. **DV:** F1 score, Exact Match (EM), retrieval precision on temporal vs. non-temporal question subsets.

H2: Temporal filtering will improve F1 on the temporal subset without degrading performance on non-temporal questions.

RQ3 *How accurately can LLMs extract temporal validity intervals from unstructured text, and what linguistic patterns most affect extraction quality?*

IV: Temporal expression type (explicit year, implicit/relative, conflicting). **DV:** Extraction precision, extraction recall, F1.

H3: Extraction precision will be highest for explicit year mentions (>80%) and lowest for conflicting date expressions.

4 Methodology

4.1 Datasets

HotpotQA. We use the distractor development set of HotpotQA (Yang et al., 2018), comprising 7,405 two-hop questions drawn from Wikipedia with ten candidate paragraphs each. Our EDA reveals that **35.0%** of questions (2,593/7,405) contain temporal expressions. Bridge-type questions exhibit higher temporal prevalence (39%) than comparison-type questions (21%). Table 1 summarizes the distribution by temporal pattern type.

MuSiQue. MuSiQue (Trivedi et al., 2022) provides 2,417 development questions with 2–4 reasoning hops, designed to preclude shortcut reasoning. **38.3%** of questions (926/2,417) contain temporal expressions. Temporal prevalence increases monotonically with hop count: 30.2% for 2-hop, 47.2% for 3-hop, and 46.7% for 4-hop questions.

Temporal Extraction Gold Standard. For RQ3, we manually annotate 200 passages sampled from the HotpotQA training set, recording ground-truth [valid_from, valid_to] for all extractable triples.

4.2 Offline Knowledge Graph Construction

Chunking. Each document is segmented into overlapping chunks of 512 tokens with a 100-token overlap, following KG²RAG (Zhu et al., 2025).

Triple Extraction with Temporal Annotation. We prompt GPT-4 to extract structured triples in the format: (head, relation, tail, valid_from, valid_to). The full prompt is in Appendix A.

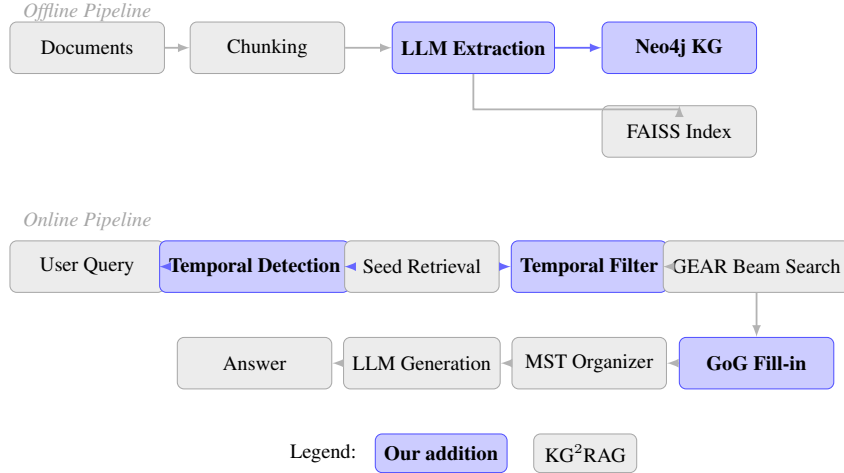


Figure 2: Overview of the TempoRAG-KG framework. Components in blue are our additions to the KG²RAG backbone. The offline pipeline constructs a temporally-annotated KG; the online pipeline applies temporal filtering before GEAR beam search and GoG fill-in for incomplete facts.

Temporal Pattern	HotpotQA	MuSiQue
Year mention	17.8%	6.7%
When/what year	7.3%	20.6%
Before/after	4.2%	7.1%
First/last/earliest	7.7%	7.1%
Time period	2.3%	3.9%
Age/duration	0.7%	1.8%
Any temporal	35.0%	38.3%

Table 1: Temporal expression prevalence in HotpotQA and MuSiQue development sets. Questions may match multiple patterns.

KG Storage and Indexing. Extracted triples are stored in Neo4j with `valid_from` and `valid_to` as edge properties. Entity names are embedded using `all-MiniLM-L6-v2` (Reimers and Gurevych, 2019) and stored in a FAISS index for efficient similarity search.

4.3 TempoRAG-KG Pipeline

The online pipeline proceeds in six steps:

(1) Temporal Year Detection. A regex-based classifier identifies whether the input query contains a temporal expression and extracts the target year y_q .

(2) Seed Retrieval. The query is embedded and the top-5 most similar chunks are retrieved from the FAISS chunk index.

(3) Temporal Filtering. For each candidate edge e with properties v_f (`valid_from`) and v_t (`valid_to`), we retain e if and only if:

$$(v_f = \text{null} \vee v_f \leq y_q) \wedge (v_t = \text{null} \vee v_t \geq y_q) \quad (1)$$

Edges without temporal metadata are retained conservatively.

(4) GEAR Beam Search. Graph expansion proceeds via beam search (Ma et al., 2025) with beam size $b = 3$ and maximum depth $d = 2$. Candidate neighbors are scored by cosine similarity to the current subgoal embedding.

(5) GoG Fill-in. When expansion reaches a dead end, we invoke a GoG-style LLM fill-in (He et al., 2024). Fill-in facts with confidence below 0.7 are discarded.

(6) MST Context Organization and Generation. Retrieved subgraph facts are organized via Maximum Spanning Tree pruning into coherent paragraphs, prepended to the generation prompt for LLaMA-3-8B or GPT-4.

4.4 Baselines

We compare against four baselines:

- **Vanilla RAG:** Dense retrieval with GPT-4; no graph.
- **GraphRAG:** Microsoft community-summary RAG (Edge et al., 2024).
- **KG²RAG:** Reproduced NAACL 2025 baseline (Zhu et al., 2025).
- **TempoRAG-KG (ours):** Full system with temporal filter + GoG.

4.5 Ablation Design

To isolate each component’s contribution (RQ2), we evaluate five conditions shown in Table 2:

Condition	Temporal	GoG	Graph Exp.
Vanilla RAG	✗	✗	✗
KG ² RAG	✗	✗	✓
+Temporal only	✓	✗	✓
+GoG only	✗	✓	✓
Full (ours)	✓	✓	✓

Table 2: Ablation conditions. ✓ = component enabled; ✗ = disabled.

4.6 Evaluation Metrics

We report:

- **F1 and EM:** Token-level F1 and Exact Match, computed overall and separately for temporal and non-temporal question subsets.
- **Retrieval Precision@5:** Fraction of retrieved chunks containing the gold supporting facts.
- **Temporal Extraction F1:** Precision and recall against gold [valid_from, valid_to] annotations with ± 1 year tolerance (RQ3).
- **Inference Time:** Average seconds per query.

5 Expected Results and Discussion

5.1 RQ1: Temporal Failure Mode Distribution

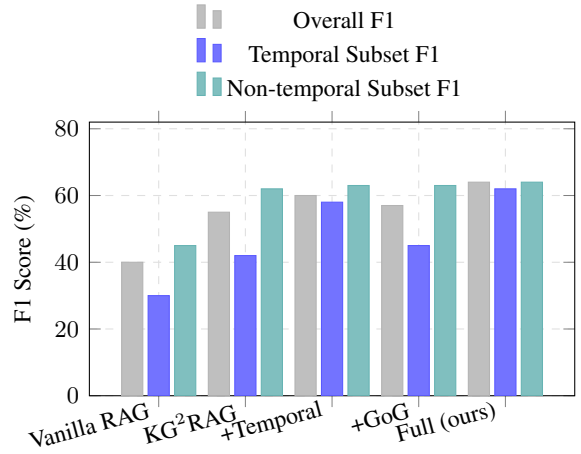
We expect stale-fact retrieval (Type 1) to constitute the majority of temporal errors in KG²RAG, as the most common temporal pattern in HotpotQA is explicit year mention (17.8%). Type 4 (temporal hop failure) is expected to be rare in HotpotQA (2-hop only) but more frequent in MuSiQue 4-hop questions.

5.2 RQ2: Impact of Temporal Tagging

We hypothesize that the full TempoRAG-KG model (Table 3) will outperform KG²RAG on the temporal question subset by a meaningful margin, while maintaining comparable performance on non-temporal questions. Table 4 shows the expected ablation structure.

5.3 RQ3: Temporal Extraction Accuracy

We expect precision $>80\%$ for explicit year-mention expressions but substantially lower recall for implicit/relative expressions. Table 5 shows the evaluation structure.



Note: Values illustrative — to be updated with experimental results.

Figure 3: Expected ablation results on HotpotQA. Gains concentrate in the temporal subset, while non-temporal F1 remains stable.

5.4 Discussion

If our hypotheses are confirmed, the results carry a broad implication for KG-RAG system design: *temporal metadata should be treated as a first-class property of KG edges, not an optional annotation.* Systems that attach validity intervals at construction time gain the ability to filter stale facts deterministically—a capability that cannot be retrofitted to static KGs without re-extraction.

The primary limitation is extraction quality on implicit temporal expressions. Our conservative fallback (retaining triples with null validity in Equation 1) mitigates but does not eliminate this risk.

6 Progress Summary

Completed.

- Literature review: 16 papers across four team members (4 papers each, per course requirements from ACL venues only).
- EDA of HotpotQA (7,405 questions: 35.0% temporal) and MuSiQue (2,417 questions: 38.3% temporal).
- Temporal failure taxonomy: four types proposed based on qualitative analysis of KG-RAG failure modes.
- Analysis of closest competitor EvoReasoner (Lin et al., 2025) with clear differentiation on three dimensions.

Model	Overall		Temporal Subset		Non-temporal Subset	
	F1	EM	F1	EM	F1	EM
Vanilla RAG	–	–	–	–	–	–
GraphRAG	–	–	–	–	–	–
KG ² RAG	–	–	–	–	–	–
TempoRAG-KG (ours)	–	–	–	–	–	–

Table 3: Main results on HotpotQA distractor dev set. Temporal and non-temporal subsets contain 2,593 and 4,812 questions respectively. **Bold** = best result. (Results to be filled after experiments.)

Condition	Overall F1	Temporal F1
Vanilla RAG	–	–
KG ² RAG (backbone)	–	–
+Temporal filter only	–	–
+GoG fill-in only	–	–
Full model (ours)	–	–
Δ vs. KG ² RAG	–	–

Table 4: Ablation scores on HotpotQA. Δ = improvement over KG²RAG backbone. (Results to be filled after experiments.)

Pattern Type	Prec.	Recall	F1
Explicit year mention	–	–	–
Implicit / relative	–	–	–
Conflicting dates	–	–	–
Overall	–	–	–

Table 5: Temporal extraction accuracy on 200 manually annotated HotpotQA passages (RQ3), with ± 1 year tolerance. (Results to be filled after annotation.)

In Progress.

- Neo4j schema design and setup with temporal edge properties.
- Temporal extraction prompt engineering and pilot evaluation on 20 HotpotQA passages.
- GitHub repository setup with README.¹

Remaining Work.

- Reproduction of KG²RAG baseline on HotpotQA distractor dev set.
- Manual annotation of 200 passages for RQ3 gold standard.
- Full pipeline implementation: temporal filter, GEAR beam search, GoG fill-in integration.
- Ablation experiments (Table 2).
- Streamlit demo with edge case scenarios.

¹Repository: [https://github.com/\[to-be-updated\]/temporag-kg](https://github.com/[to-be-updated]/temporag-kg)

Potential Concerns. The primary risk is temporal extraction quality on implicit expressions. If extraction F1 falls below 60%, the temporal filter may introduce false negatives. The conservative null-retention strategy in Equation 1 ensures graceful degradation.

7 Conclusion

We have presented TempoRAG-KG, a temporally-aware Knowledge Graph-augmented RAG framework that addresses a systematic and underexplored failure mode in existing KG-RAG systems: the inability to distinguish facts valid at different points in time. By attaching explicit validity intervals to KG triples and filtering graph expansion accordingly, TempoRAG-KG targets the 35% of HotpotQA and 38% of MuSiQue questions that require temporal reasoning. Our temporal failure taxonomy provides the diagnostic foundation needed to understand when and why temporal reasoning fails in graph-augmented systems, and our planned extraction study characterizes the practical accuracy ceiling of LLM-based temporal annotation.

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A Temporal Extraction Prompt

You are a knowledge extraction assistant. Given the following text passage, extract all factual triples. For each triple, also infer temporal validity if mentioned or implied.

Format each triple as JSON:

```
{
  "head": "entity name",
  "relation": "relation type",
  "tail": "entity or value",
  "valid_from": year (int) or null,
  "valid_to": year (int) or null,
  "confidence": float between 0 and 1
}
```

Rules:

- Only extract explicitly stated facts
- For valid_from/valid_to: use null if unknown
- If ongoing, set valid_to to null
- Conflicting dates: extract as separate triples

Text: {chunk_text}

B EDA Methodology

Temporal questions were identified using seven regex patterns applied to the lowercased question string. A question was classified as temporal if it matched at least one pattern. The patterns target: (1) explicit year mentions, (2) when/what-year phrases, (3) before/after/since, (4) time period expressions, (5) first/last/earliest superlatives, (6) age and duration phrases, and (7) temporal relation words (predecessor, successor, etc.). The EDA script is available in the project repository.